Australian
National
University

12 January 2026
Ian G. Brennan
Research School of Biology
+61 0437 208 347
ian.brennan@anu.edu.au

12 January 2026

Dear Prof. Akira S Mori,

We would like to begin by thanking both you and the two anonymous reviewers for your creative and valuable thoughts. On behalf of myself and my coauthors, we appreciate your time and contributions.

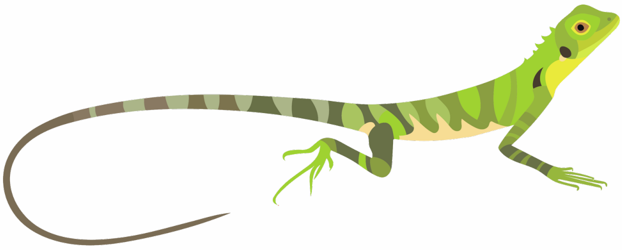
All reviewers provided carefully considered comments and we have done our best to accommodate those suggestions here. The most significant changes include a clearly stated and testable hypothesis in the introduction, clearer definition of generalism/specialization, and an extensive niche modelling exercise to test our hypothesis. We believe these additions substantially improve our argument and the quality of the manuscript.

Below please find reviewer comments in green, and our responses in black. New text added to the manuscript is highlighted. We have formatted a page break between each reviewer. Where necessary, we have interjected our responses among reviewer comments.

On behalf of myself and coauthors, we thank you for your time and consideration,

Dr. Ian G. Brennan

Postdoctoral Researcher
Division of Ecology and Evolution
Research School of Biology
The Australian National University, Canberra, ACT
ian.brennan@anu.edu.au / iangbrennan@gmail.com



Referee 1:

The manuscript contains an extensive morphological and phylogenetic analysis of dragon lizards, examining an interesting idea surrounding how evolution through morphospace is likely to occur, with the implicit hypothesis that specifically the center of morphospace acts as a sort of hub, from which more extreme forms are generated (rather than extreme forms leading directly to other extreme forms). The main problem with the manuscript as currently presented is that this idea is not really clearly stated in the introduction, nor is it directly tested in the results—it is only circled around to in the discussion using qualitative rather than analytical quantitative support from the dataset.

This is an important comment regarding how our manuscript was previously written. To better communicate our hypothesis we have added the following sentences to the introduction (Lines 78-83):

> The complex topography of the landscape, including holes, dictates viable paths from existing morphologies into new ones. As such, the ideas of adaptive radiation and ecological opportunity have long been entwined with Simpson's view of movement across the landscape (Schluter 2000; Simpson 1944; Yoder et al. 2010). However, we still do not understand how new peaks form in disparate regions of the adaptive landscape. One testable hypothesis is that the center of morphological space acts as a hub or bridge leading to more distant adaptive zones (Dennis et al. 2011; Futuyma & Moreno, 1988). The expectation being that the morphological center is occupied by taxa with generalist morphologies that enable broad niche breadths, and this facilitates evolutionary movement into narrower specialist niches.

The manuscript is further led astray by substantial discussion of "generalists", though generalism is never clearly defined, and generalization is not quantified in the manuscript. This makes all the claims about the importance of generalists come across as extremely handwavy. Specifically, it seems like the authors are only focused on habitat use generalization along the terrestrial to arboreal axis, for which they are using morphology as a proxy (despite discussing why this can not really be done multiple times in the manuscript), and then conflating niche placement with niche breadth (generalization). Namely that semi-arboreal species (which I interpret to mean species that tend to occupy low vegetation, with occasional forays to the ground to acquire food), are generalists, where as terrestrial species and arboreal species are specialists. However, without actually characterizing breadth of habitat use there is no ability to make this claim (or more importantly to falsify it). In other words, for a semi arboreal species to be a generalist we need to have a niche axis (e.g. perch height, or perch stratum or perch substrate types) and show that the generalist uses more heights/strata than either arboreal species or terrestrial ones. Instead it just seems that all we can say is that semi-arboreal lizards are intermediate in their perch height use between arboreal and terrestrial ones, but can make no statement about generalization per se. Fortunately, I don't think generalization per se is necessary to test the central idea (i.e. that the center of morphological space acts as an evolutionary hub for generating morphological biodiversity), and so all talk of generalization can (and should) be struck from the manuscript (unless the authors actually quantify generalization so as to be able to make rigorous statements about it. There will however need to also be rigorous analytical tests examining whether evolutionary moves through morphospace are more likely to go first to the center, prior to branching out to extremes. This is implied in the left panel of figure 4, but quantitative backing is needed through null modelling or other approaches.

We appreciate your very careful consideration and creative thoughts about how generalism can be looked at more carefully in our study. We agree with you that we were ambiguous in how we defined “generalism” and in the original version of our manuscript we did not use quantitative measures to assess this. To address this point we have clarified our thoughts on generalism/specialization and undertaken additional analyses to test the ideas presented in our paper. Specifically, we define generalism through the lens of niche breadth by characterizing substrate/habitat usage across species. We then model the evolution of niche breadth to investigate the idea that specialists (narrow niches) evolve from generalists (wide niches) and not from other specialists. These ideas and methods are now outlined on Lines 181-201:

> *Defining Generalism through Habitat Use*
> To address the concepts of generalism and specialization, we focused on niche breadth. Specialization can occur on varied axes, with a given taxon being a generalist on one (e.g. habitat type), but a specialist on another (e.g. diet) (Poisot et al. 2011), like the thorny devil Moloch horridus that occurs throughout arid Australia but exclusively eats ants. We focus on the fundamental Eltonian niche of species morphologies and the realized niche through substrate usage (Devictor et al. 2010). We first discretized surface features and microhabitats used by dragon lizards into six categories which have different functional demands: aquatic, rock, terrestrial, low structure, high structure, and arboreal (see Supplement for details). We scored all species for these traits and treated the trait matrix in two ways. First, we generalized niche breadth by summing trait usage. Values of 1-6 indicate highly specialized (low values) to absolute generalist species (high values). Second, to investigate transitions among specialist and generalist states we classified taxa as either specialist (one or two substrates used) or generalist (>=3 substrates used) category. For both character codings we then fitted a series of Markov models using phytools (Revell 2024) to compare theoretical pathways for the evolution of niche breadth and specialization in dragons. These included equal and independent transition rates among states, stepwise (increasing or decreasing breadth by one state) and loose stepwise models (multiple states can be gained or lost), and a strict model of increasing specialization. Importantly, we applied a derivative in which niche breadth acquisition and loss are allowed separate transition rates (see Supplement for more details) to all eligible models.

In the absence of a clearly stated and falsifiable hypothesis in the introduction, and rigorous approaches to test that hypothesis in the results the manuscript as currently presented comes off as extremely descriptive. Despite this fairly negative review for the two specific reasons mentioned above, I do want to stress that I think this dataset is absolutely amazing, and the work put into the analysis, and the figure generation surrounding it is top notch. The manuscript (as currently written, in my mind) simply lacks the needed focus to make a substantial contribution to the fields of ecology and evolution. However, I think this dataset absolutely can make that important and general contribution with some relatively minor tweaks. But otherwise the more descriptive approach taken would likely be a better fit at more specialized and less conceptual journals.

We appreciate the enthusiastic comments about the dataset, analyses, and figures. As mentioned above, we have addressed the main critique - that the presentation is overly descriptive - by incorporating comparative modelling to support our hypotheses about the evolution of niche breadth and morphology. We have worked to align our project more closely with the expectations of Ecology Letters.

Line comments:

L11: Unclear how “generalist forms” are defined. Different fields within ecology and evolutionary biology have different definitions of “generalist” so it’s crucial to define.

We agree with your statement about the ambiguity of the term “generalist”. It’s difficult to incorporate this nuance into a 150 word abstract (or title). To address this, we have changed the phrasing to “generalist habitat use”. We have also changed the following sentence to better reflect our methods: “Ancestral dragons were likely arboreal, but species with broad niche breadths and intermediate morphologies provided pathways to subsequent specialization.”

L20: I’ve always thought that the landscape included the peaks (i.e. the fitness associated with phenotypes) where as morphospace just delimited the territory (with no claims about fitness). Is this not the case? Be clearer in how you’re defining and equating landscape vs morphospace.

This is a really interesting topic and one we wish we had more space to address. Because the adaptive landscape is a metaphor it is inherently thought of differently by different people. However, a landscape can be defined by any measured traits. Here we exclusively use morphology, and so the landscape is equivalent to the morphospace. This use follows the design by Sewall Wright who defined the landscape by gene combinations to make an equivalent gene-space (e.g. Fig. 2 of Wright 1932). We admit that we do not have the ability to make claims about fitness in our morphospace, but other instances do (e.g. Stroud et al. 2023 https://doi.org/10.1073/pnas.222207112). In our case, we consider peaks as populated regions of the morphospace, particularly where there may be aggregations of multiple species.

L57 (L77): Again, unclear what you mean specifically by “generalist”.

This phrasing was unintentionally vague. We have adjusted it to read:

> Lastly, we identify that the diversity of forms of Australian dragons likely evolved from ancestors with broad niches that filled space in the adaptive morphological surface and acted as intermediates connecting an otherwise patchy landscape.

Introduction generally: I had trouble finding the hypothesis/theory being tested.

We hope that the additions mentioned above (see response to first comment) better outline the hypothesis tested in our work.

~L180: This approach (looking at the hypervolume) assumes no extinction. Can you comment here on how realistic you think this is, and how much it matters for the conclusions of the analysis, and the specific questions you are addressing. [And readdress during discussion]

All analyses that lack fossil information are sensitive to assumptions regarding extinction. In this particular case, we are looking at the accumulation of phenotypic volume. Because volume is dictated by (and sensitive to) trait combinations at the fringes of known space, it is potentially less sensitive to bias than other measures like density. If our expectations about the evolution of morphological diversity in dragons are true (ancestrally intermediate morphologies which lead towards young derived morphologies), then we would not expect extinction to considerably affect our conclusions about the accumulation of morphological volume. To address this idea we have added text to the discussion:

> The morphological evolution of dragon lizards has been a heterogeneous process, varying temporally, phylogenetically, and across traits. Broadly, dragons have gone through periods of early (30-25 my) and late expansion (5-0 my) in the morphological hypervolume, bookending a period of little innovation during much of the Miocene (Fig.S8). Admittedly, this pattern lacks information provided by extinct species. However, if our expectation that ancestrally intermediate morphologies yield young derived morphologies is true, then we would not expect extinction to considerably affect our conclusions about the accumulation of morphological volume. This does establish further questions about how biased extinction of morphologically extreme species may drive the pattern that we see.

L196: Is the 'a' in parantheses supposed to be an alpha?

Yes. We have corrected the 'a' to be an α.

L120 ~L240: It wasn't tremendously clear how time calibration was achieved and slightly more information about this could be provided in the methods (around L120, rather than just saying to look in the sup mat). Nor was it clear exactly how seriously we should take the estimates of events in these paragraphs. Confidence intervals would be helpful around these dates. Also the bit about <600K years, pointing to figure 1, is confusing, because it doesn't look like any splits are happening in the <1million year timeframe. Please clarify.

Thank you for your interest in the calibration methods, unfortunately we are limited somewhat by space (<5k words) and so have to keep the specifics of the fossil choices and applications in the supplement. We have, however, tried to include a discussion of the placement and importance of these fossils (see Supplement Node Priors and Agamids in the Fossil Record). With regards to the comment about <600k years, we apologise for the lack of clarity. Our intention was to highlight a window of 600k years (less than a million year period) at the base of the amphibolurine tree. To clarify this, we have adjusted the text to read:

> Rapid speciation events at the base of the amphibolurine tree obscure some relationships. This is likely the result of quick successive splits that occurred in a narrow temporal window, <600 ky (Fig.1).

L255: The topics discussed in the following lines seem like they warrant a main text figure to me, to be able to easily evaluate these claims (or be integrated into figure 1).

We have incorporated the volume-through-time plot that shows the expansion of the morphological hypervolume as an addition to Figure 1. We compare the inferred empirical pattern against data simulated under constant rate Brownian Motion as a null model.

L263: Again, what do you actually mean by “generalist”. Generalism now seems to be specifically about habitat use rather than to be anything about generalization more broadly, or associated with morphology per se. This was not clear from the set up in the introduction. There’s obviously a link between habitat use (and also likely habitat use breadth i.e. generalization) and morphology, but none of the analyses are actually testing that, because no independent data on ecological generalization is being brought to bear. It’s crucial to highlight that generalism typically acts independently along niche axes, and so if habitat generalization is specifically what’s being considered here it’s crucial to specify that. Otherwise it gets conflated with things like diet breadth, or thermal tolerance, or macrohabitat breadth etc which are outside the purview of what is (seemingly) able to be assessed here. Though honestly, “generalism” is beginning to come across as a meaningless “other” category that can’t be actually defined, especially in the absence of ecological data quantifying to what degree things are “arboreal”, “terrestrial”, or “generalist”.

Again, this is a fair criticism which we believe we address throughout our revised manuscript. To avoid conflating morphological regime assignments and habitat use niche assignments we have removed the reference linking morphological regimes to habitat use in this section. Instead we refer to groups by their inferred regime following Fig.3.

L266: The sudden talk here of “generalists” being “semi-arboreal” raises the question as to whether niche center/position is being conflated with niche breadth.

We have amended this passage to avoid blending the ideas of niche breadth and/or habitat use with morphology.

L275: This would all be stronger if the ancestral resconstruction was done on actual habitat use, rather than making ecological claims based solely on morphology and calling a morphological group “arboreal-type”. Some of this is just naming conventions. This would be less problematic if you just named the groups something neutral (like R1 in the figure) and said something like “RandomForest predictions suggest an ancestor from R1 (which is made up of primarily arboreal species in the present day) for the Amphibolurinae” or something of the sort to make it very clear that the inference about ecology is a few steps removed from the data at hand. (Or just analyse the ACE for the habitat data, and be able to be a bit more firm about this supposition if both morphology and ecology are pointing in the same direction).

We appreciate your recommendations. To address this comment we have expanded our modelling approach and incorporated data that summarizes habitat use. We present this first in the Materials and Methods section *Defining Generalism through Habitat Use*:

> To address the concepts of generalism and specialization, we focused on the idea of niche breadth. Specialization can occur on varied axes, with a given taxon being a generalist in one (e.g. habitat type), but a specialist on another (e.g. diet), like the thorny devil *Moloch horridus* (Poisot et al. 2011). We focus on the fundamental Eltonian niche of species morphologies and the realized niche through substrate usage (Devictor et al. 2010). We started by discretizing surface features and microhabitats used by dragon lizards into six categories which have different functional demands: aquatic, rock, terrestrial, low structure, high structure, and tree (see Supplement for details). We scored all species for these traits and treated the trait matrix in two ways. First, we generalized niche breadth by summing trait usage, resulting in possible values of 1-6 indicating highly specialized (low values) to absolute generalist species (high values). Secondly, to investigate transitions among specialist and generalist states we classified taxa as either a specialist (one or two substrates used) or generalist (>=3 substrates used) category. For both character codings we then fit a series of Markov models using *phytools* (Revell 2024) to compare theoretical pathways for the evolution of niche breadth and specialization in dragons. These included equal and independent transition rates among states, stepwise (increasing or decreasing breadth by one state) and loose stepwise models (multiple states can be gained or lost), and a strict model of increasing specialization. Importantly, to all eligible models, we applied a derivative in which niche breadth acquisition and loss are allowed separate transition rates (see Supplement for more details).

We summarize the outcomes of this exercise in the Results section *Evolution of Niche Breadth*:

> Discrete trait model fitting of niche breadth favored a model in which transitions occur through neighboring (n±1) or near-neighboring (n±2) states and show a faster transition rate towards decreasing niche breadth (increasing specialization) (Fig.SX ModelFits). Model fitting of transitions among specialist (1 substrate) and generalist (3+ substrates) habitat usages favored a model where transition rates are faster between generalist and specialist states, and slower—but not absent—among specialist states. (Fig.Y)

And discuss this briefly in the discussion section *Mapping Morphology to Ecology*:

> Modelling niche breadth supports this by preferring pathways of narrowing substrate usage and favoring transitions between specialist and generalist states over transitions from one specialist to another (Fig.2).

Further, we visualize the results of these model fitting exercises in a new Figure 2 which explores the evolution of niche breadth and substrate use along the phylogeny. We believe this improves our ability to link morphological regimes and niche breadth and strengthens our argument for broad breadth in the niche of ancestral Australian dragons.

L382: This is not convergence on a lifestyle—as the previous sentences were outlining how ecologically these species/groups have very different lifestyles. Any perceived convergence is due to the artificial lumping in an “arboreal” category, when the actual species are specializing to different niches within the arboreal zone. Mahler et al 2013 Science

(Exceptional Convergence on the Macroevolutionary Landscape in Island Lizard Radiations) seems like a highly relevant source to think about here and elsewhere in the discussion.

This is another interesting idea about how we discretize ecologies, particularly how finely we cluster them. We agree that these species are "specializing to different niches within the arboreal zone", however, categorizing them together as arboreal is still a useful action. All categorizations rely on some semantics, and here we use "aboreal" to differentiate from things that are terrestrial, fossorial, or any other broad habitat category. Our intention is not to make the case that "aboreal" (or any other category) is a singular ecology, but instead (like you point out) to highlight the fact that there are ways to specialize *within* these categorizations. To address your comment, we have rephrased it to read "In this situation, convergence on an arboreal lifestyle is met with only incomplete morphological similarity (Grossnickle et al. 2024)."

L415: To me this seems to be the main idea that may be novel: "In this way, generalists negate the need to cross wide fitness valleys in the adaptive landscape. Instead, species can explore across ridges or broad fitness peaks to reach distant morphospaces. This provides more evidence that transitions across the landscape from specialist to specialist are uncommon and likely require an intermediate form.". The problem is that generalism is not actually measured, but is more assumed as 'whatever is at the center of morphospace'. Whether the center of morphospace actually corresponds to true generalist is an interesting question, but would need to be quantified specifically based on habitat use data (or whatever niche axis generalism is being proposed along). That things do not jump all the way across morphospace is not surprising, though the idea that things will shuttle specifically to the center, before then going back to alternative extremes is an interesting idea (i.e. rather than skirting the edges). But the analytical method to test that idea per se would need to be conducted with some null models, and currently the methods undertaken do not rigorously assess this idea.

We agree that this is an important result and have even highlighted it in our abstract ("Ancestral dragons were likely arboreal, but species with broad niche breadths and intermediate morphologies provided pathways to subsequent specialization"). As mentioned previously, we have attempted to address the criticism that we did not previously measure generalism or model the actual pathways. A more thorough explanation of what we carried out is provided above in response to the comment on "L275: This would ..". It is difficult to model this explicitly using morphological data because the morphological centroid is biased by imbalanced species richness across ecologies. High numbers of *Ctenophorus* and *Tympanocryptis* pull the morphological centroid towards these morphotypes, many of which are highly specialized and show narrow niche breadths. Instead, we scored species for habitat use and used this to summarize species niche breadth. We then fit Markov models which tested the idea of transitions between low and high niche breadth states to determine the most likely path of evolution. We summarized results for ancestral nodes under the preferred models. Results support the idea of biased transition between specialist (low niche breadth) and generalist (high niche breadth) states, with limited transition between specialist states. These results are presented visually in Figure 2 and Figure S13. This is an appropriate method for addressing the question.

Fig 1: I would request a lineage through time plot as an inset somewhere here, to get a clearer image of what you talk about in the figure legend (e.g. burst of diversification in the Oligocene).

Include 95% confidence ellipse (or SE ellipse) on root state. At present it's hard whether to know how seriously to take the placement of the triangle.
Figure legend says that white circles on the tree represent PP<0.95. But I don't see any white circles. Did I just miss them because they are so few? Make more obvious.

Regrettably the tree in Figure 1 did not have any white circles, we are not sure how they were omitted. There are very few (3) and they are located in the *Diporiphora bilineata*, *Diporiphora nobbi*, and *Ctenophorus fordi* clades. We have chosen not to include a lineage through time plot because the pace of diversification of the group is not of direct interest to this manuscript, and so have removed the text that mentioned this. However, we have retained the LTT plot that was presented in Fig.S8.

Fig 2: For the sake of the reader, and since the interpretation primarily revolves around habitat use (using terrestrial, semi-arboreal (generalist?) and arboreal as the categories), it would be helpful to make the color coding more clearly correspond to the habitat use. I.e. currently both orange and blue are (primarily) terrestrial species groups. It would therefore be helpful to make them the same type of color (i.e. cool colors), which are spectrally most distinct from the habitat type they most contrast with (arboreal). This will minimize cognitive dissonance when looking at this figure, and make the conclusions easier to track.

At the request of the reviewer, we have adjusted the colors to more accurately reflect the regimes they match to. A re-colored version is included as Figure 3.

Referee 2:

Comments for the Authors

I very much enjoyed reading this manuscript. The article is well written and clear, the methodology was well articulated, the figures are beautiful, and the main messages were compelling. I have a couple major comments, which are really more like ideas I had that I think would strengthen the messages they accompany, and otherwise I list some minor comments along with line numbers.

We appreciate your enthusiasm for our manuscript and hope readers of Ecology Letters enjoy it similarly.

Major comments

A major point made in this manuscript is that generalist morphologies provide a way to shift between more extreme morphologies – that you don't really see a switch from one morphological specialization to another. I think this is a very interesting finding, and the authors make a compelling case, but it seems to me that there is an opportunity to provide quantitative support for this as well, and that would strengthen the argument made here. Through Mk or SSE models, one could compare the transition rate between specialist forms and generalist-specialist pairs. In fact, doing so would provide a quantitative and highly complementary analysis that would provide the authors with model-based results to back up one of the main points they make in this manuscript. And if the authors are worried that SSE or Mk models do not sufficiently accommodate the trait rate heterogeneity that might be in this data (and as such would not mirror the Fabric model), if the authors have used RF to assign all internal nodes to regimes, seems like a transition rate could be calculated directly.

One analysis that I thought was quite interesting was the calculation of a morphological hypervolume at 0.5 million year time intervals across the tree. What I felt was missing though was an understanding of whether or not the trajectory shown in figure S8 was similar or different than should be expected under some null model. It seems to me that it wouldn't be too complicated to simulate data based on fitted values from the empirical data (under a global Brownian motion model) and generate a null distribution of hypervolumes over time, much like people generate Brownian motion expectations for disparity through time plots. This would enable the authors to point to time periods when morpho-volume was increasing at elevated rates, or where morpho-volume was unusually static.

We agree that the idea of designing and fitting Markov models to determine the pathways of transitions between generalist and specialist forms is a great idea. To implement this, we have scored species for habitat and substrate use and used that to approximate niche breadth. We then modelled the evolution of niche breadth two ways and present the results in Figure 2. For more information about our approach, please see our response to Reviewer 1 above.

Minor comments

Line 44-45: Opportunity is listed here alongside ecology, habitat and geographic range. But opportunity strikes me as something that is fundamentally different – it might be unoccupied ecological space or unsaturated habitat. So it seems that opportunity should be treated differently in this sentence.

This is a good point, opportunity is not a measurable axis the way ecology, habitat, and geographic range are. We would love to explore the idea further, but are constrained by space, and so have removed “opportunity” from this statement.

Line 180-183: If I am understanding correctly, for every 0.5 million years, the authors took all estimated ancestral states of species from the root to that 0.5 million year increment, and calculated the volume and density of the morphospace occupied? If so, then the volume can only increase with time, as more species are added and never removed, correct? Would be helpful to explain a touch more in the text.

Our wording was slightly unclear. At 0.5 million year increments we collected trait values for all contemporaneous lineages. This required extrapolating trait values linearly along branches given start and end values at nodes and a constant evolutionary rate for each branch. We did this from the root to tips at 0.5 million year timeslices, and then fit a hypervolume and estimated the volume. We did not fit the hypervolume from the root to each 0.5 million year increment because that would lump together non contemporaneous lineages, or multiple measurements for the same lineage. If that were the case, yes, you would only see an increase in volume through time or stasis, never declining. Instead, we have included the volume-through-time plot as (C) in Figure 1, and you can see there are dips and pauses in the trajectory. This is a result of the greater variability in the way traits are allowed to evolve under the *fabric* model. We plot our empirical trend against Brownian null expectations, as requested. Admittedly, this pattern is likely less variable than if we had well-sampled fossil information and extinct taxa to inform ancestral states and trends in deep time. We have amended the text to better explain our method:

> To track the evolution of dragon morphospace through time we extrapolated trait values linearly along branches at 0.5 million year intervals given start and end values at nodes and constant estimated evolutionary rates. We then reduced observed and estimated ancestral values to three principal component axes and fit multidimensional hypervolumes using *hypervolume* (Blonder et al. 2018) to extract the volume.

Line 190: Did the authors use the scalarOU model within PhylogeneticEM?

Yes we specified ‘process = “scOU”’ in our PhylogeneticEM analyses. We have made this information available in the R code found in Scripts/Morphological_Analysis/03_Regimes.R at the GitHub repository https://github.com/IanGBrennan/Amphibolurinae.

Lines 198-207: This is a very interesting approach, of using RandomForest to predict which regime ancestral taxa may have belonged to based on the estimated ancestral morphological reconstructions. I don’t see citations here – to the authors’ knowledge, is this the first application of such an approach in this context?

We appreciate your interest in this method. We are not familiar with other studies that have used this approach for ancestral taxa. It’s doubtful that we’re the first, however, as conceptually similar methods have been applied in archaeology, paleontology, and even genomics:

Hefner, J.T., Spradley, M.K. and Anderson, B. (2014), Ancestry Assessment Using Random Forest Modeling. J Forensic Sci, 59: 583-589.
Cole, K. E., Yaworsky, P. M., & Hart, I. A. (2022). Evaluating statistical models for establishing morphometric taxonomic identifications and a new approach using Random Forest. *Journal of Archaeological Science*, *143*, 105610.
Hallee, L., Khomtchouk, B.B. Machine learning classifiers predict key genomic and evolutionary traits across the kingdoms of life. Sci Rep 13, 2088 (2023).

Line 398 (L389): Fig.SX12? Probably meant to remove the X.

Thanks for catching this. Removed.

Line 414: check parenthesis and comma placements

Thank you. We have moved the reference to Fig.4 to the end of the sentence to avoid the awkward commas.

I very much appreciated the clarity and level of detail in the methods section.

We appreciate this acknowledgment. It is hard to strike a balance between brevity and accuracy while still supporting reproducibility. We hope that any elements not included in our methods section will be readily understandable in our commented code.

Beautiful figures.

Thanks! Visualizations are an important part of communicating complex ideas, so we spend a lot of time and energy on them. We appreciate the kind words.

Figure 1: Legend states that white circles denote PP < 0.95, but I don't think there are any white circles.

Yes, we somehow failed to add the white circles. There are only three, and they have been added back to the plot.

Figure 1: Could you plot the theoretical OU optima in the morphospace plot?

This is a good question and thought. Unfortunately, extracting the theoretical optima from PhylogeneticEM is not as easy as it seems because of the way regimes are fit. If we assume that the optima are the root trait values at shift points, then yes, it is feasible to extract those values by matching the position of regime shifts (branch number) to estimated ancestral trait values derived from the function *imputed_traits()*. This allows us to determine that the optimal trait values are reasonable, and we have inspected these values. However, we do not think that adding these values is likely to provide meaningful information that cannot already be extracted from the functional space plot in Figure 1B.

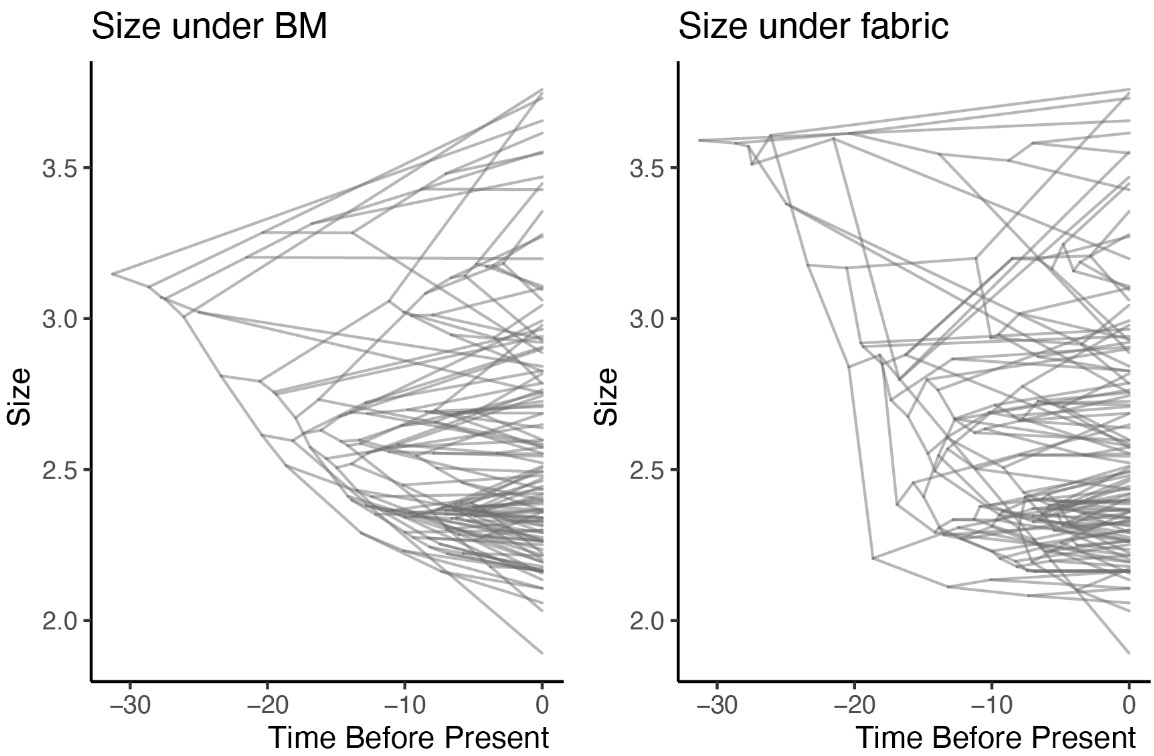
Editor:

Editors Comments for the Author(s):
This was an interesting paper to read, but somewhat surprising to get at ELE. The dataset is very impressive. Much of the paper is, however, a very sophisticated phylogenetic analysis rather than a test of ecological theory. Though couched in the language of "peaks in the adaptive landscapes" (clumps of species in morphological space, corresponding to general modes of life (arboreal, terrestrial, generalist, etc.)) the main conclusion and the paper's title -- that the specialist lineages in Australia had a generalist ancestor, even though the invading lineage might not have been a generalist -- does not yet rest on any clear statistical test. The pattern is of an arboreal lineage invading Australia, followed by a transition to a generalist morphology, with several specialist morphologies evolving subsequently (each representing a peak in the current adaptive landscape). At its core, this is still a story based on reconstructed PCA scores using mixed BM/OU-models. Note, I have no initial issue with the extensive dataset nor how reconstructed lineages are characterized or mapped onto the tree; the mapping seems state-of-the-art in a first reading.

We appreciate your interest in our manuscript. To better test our hypothesis about how generalist ecologies may facilitate movement across the adaptive landscape we have carried out more explicit model fitting analyses. Following the valuable advice of two fantastic reviewers, we have more accurately addressed the idea generalism/specialism through a habitat use and niche breadth lens. This required scoring all species for habitat use to serve as a proxy for niche breadth and model the transitions among states. We explain our methodology best in the text and in response to Reviewer 1. To summarize our results, we find evidence that the ancestral Australian dragon likely had moderate-to-high niche breadth and used multiple (3 or more) substrate types. Importantly, we find that transitions between generalist (high niche breadth) and specialist (low niche breadth) states are more likely than among specialist states, supporting our idea that wide niche breadth facilitates transition among ecologies and promote specialization. We believe that these additional methods provide an important quantitative support for our ideas that may not have been present in our initial submission.

Both reviewers agreed that tests of the reconstructed scenario are needed. As the authors well-know, internal nodes that lead to disparate tips will have somewhat "generalised" reconstructions under most stochastic models due to a regression to the mean. The invading lineage is reconstructed as a non-specialist (arboreal), but with what must be considered low confidence (60-70%), and even this is muddied by the fact that two of the species that look like it are aquatic. Overall, this key ancestor most resembles the lineages that are closest to it (as expected), with the exception of Moloch (here the multi-optimum OU model shines).

We agree that without internal node information (e.g. from fossils), all reconstruction methods result in somewhat “generalised” estimates for internal nodes. This result, however, is more pronounced under some models than others, as expected under the model assumptions i.e. Brownian Motion. The flexible and more dynamic nature of the *fabric* model, which allows branch specific evolutionary rates and/or directed evolutionary trajectories, results in less “generalized” reconstructions. These reconstructions still lack a confident empirical basis such as fossils, but are estimated under a model that fits the observed data far better for most traits (see Results). Below we visualize the difference in ancestral reconstructions under two competing models, the BM null and the preferred *fabric* model, for the "size" trait.



With regards to the uncertain ecology of the ancestral amphibolurine and Australian dragons, we agree that there are some important caveats. We attempt to be as transparent about the uncertainty as possible, principally by visualizing the support for competing regime assignments in Figure 3. Additionally, our niche breadth and substrate usage modelling provides further nuance to the story. Results from model fitting suggests the ancestral amphibolurine likely had a narrow niche breadth (50% weight for 2; >75% weight on trees) and the ancestral Australian dragon likely had a broader niche breadth (60% weight for 3; 70% multiple substrates). These conclusions are based on the data we have collected, and we look forward to better evidence from fossil data.

As far as the position, morphology, and habits of *Intellagama* and *Physignathus*, this is a very interesting case that we wish we had more space to devote to. Phenotypically, these species are tree dragons that have been placed horizontally (instead of vertically) in riparian habitats. They spend more time on the ground than either the *Hypsilurus* or *Lophosaurus* tree dragons, but are highly adept climbers, and active swimmers. Classifying these two "water dragons" presents an interesting test case of our methods.

I do not mean to suggest that there are artefacts at work here, but only to say that reconstructing ancestral morphologies is fraught, and so clear alternative, competing, models are necessary.

Please see our answer above and response to Reviewer 1 as evidence of our approach to more clearly modelling the evolutionary processes through competing statistical models. We do recognize difficulties in reconstructing ancestral morphologies, but we believe the methods we propose are an improvement over the standard methodology.

I will end by agreeing with the reviewers that the figures are extremely compelling. Whoever did them has an artistic and didactic eye as well as clear technological prowess.

As we mentioned in response to Reviewer 2, we strongly believe in the importance of figures as visual tools for communicating complex concepts. We hope they improve the value of our manuscript and further engage our readers.